

OpenStax Calculus Volume 2

Section 7.3 Exercises: Polar Coordinates

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In the following exercises, plot the point whose polar coordinates are given by first constructing the angle θ and then marking off the distance r along the ray.

1. $(3, \frac{\pi}{6})$
2. $(-2, \frac{5\pi}{3})$
3. $(0, \frac{7\pi}{6})$
4. $(-4, \frac{3\pi}{4})$
5. $(1, \frac{\pi}{4})$
6. $(2, \frac{5\pi}{6})$
7. $(1, \frac{\pi}{2})$

For the following exercises, consider the polar graph below. Give two sets of polar coordinates for each point.

[Figure: The polar coordinate plane is divided into 12 sectors. Point A is drawn on the first circle on the first spoke above the $\theta = 0$ line in the first quadrant. Point B is drawn in the fourth quadrant on the third circle and the second spoke below the $\theta = 0$ line. Point C is drawn on the $\theta = \pi$ line on the third circle. Point D is drawn on the fourth circle on the first spoke below the $\theta = \pi$ line.]

8. Coordinates of point A.
9. Coordinates of point B.
10. Coordinates of point C.
11. Coordinates of point D.

For the following exercises, the rectangular coordinates of a point are given. Find two sets of polar coordinates for the point with angles in $(0, 2\pi]$. Round to three decimal places.

12. $(2, 2)$
13. $(3, -4)$
14. $(8, 15)$

- 15. $(-6, 8)$
- 16. $(4, 3)$
- 17. $(3, -\sqrt{3})$

For the following exercises, find rectangular coordinates for the given point in polar coordinates.

- 18. $(2, \frac{5\pi}{4})$
- 19. $(-2, \frac{\pi}{6})$
- 20. $(5, \frac{\pi}{3})$
- 21. $(1, \frac{7\pi}{6})$
- 22. $(-3, \frac{3\pi}{4})$
- 23. $(0, \frac{\pi}{2})$
- 24. $(-4.5, 6.5)$

For the following exercises, determine whether the graph of each polar equation is symmetric with respect to the x -axis, the y -axis, or the origin.

- 25. $r = 3 \sin(2\theta)$
- 26. $r^2 = 9 \cos \theta$
- 27. $r = \cos(\frac{\theta}{5})$
- 28. $r = 2 \sec \theta$
- 29. $r = 1 + \cos \theta$

For the following exercises, describe the graph of each polar equation. Confirm each description by converting into a rectangular equation.

- 30. $r = 3$
- 31. $\theta = \frac{\pi}{4}$
- 32. $r = \sec \theta$
- 33. $r = \csc \theta$

For the following exercises, convert the rectangular equation to polar form and sketch its graph.

- 34. $x^2 + y^2 = 16$
- 35. $x^2 - y^2 = 16$
- 36. $x = 8$

For the following exercises, convert the rectangular equation to polar form and sketch its graph.

- 37. $3x - y = 2$
- 38. $y^2 = 4x$

For the following exercises, convert the polar equation to rectangular form and sketch its graph.

39. $r = 4 \sin \theta$

40. $r = 6 \cos \theta$

41. $r = \theta$

42. $r = \cot \theta \csc \theta$

For the following exercises, sketch a graph of the polar equation and identify any symmetry.

43. $r = 1 + \sin \theta$

44. $r = 3 - 2 \cos \theta$

45. $r = 2 - 2 \sin \theta$

46. $r = 5 - 4 \sin \theta$

47. $r = 3 \cos(2\theta)$

48. $r = 3 \sin(2\theta)$

49. $r = 2 \cos(3\theta)$

50. $r = 3 \cos(\frac{\theta}{2})$

51. $r^2 = 4 \cos(2\theta)$

52. $r^2 = 4 \sin \theta$

53. $r = 2\theta$

54. [T] The graph of $r = 2 \cos(2\theta) \sec(\theta)$ is called a strophoid. Use a graphing utility to sketch the graph and, from the graph, determine the asymptote.

55. [T] Use a graphing utility and sketch the graph of $r = \frac{6}{2 \sin \theta - 3 \cos \theta}$.

56. [T] Use a graphing utility to graph $r = \frac{1}{1 - \cos \theta}$.

57. [T] Use technology to graph $r = e^{\sin(\theta)} - 2 \cos(4\theta)$.

58. [T] Use technology to plot $r = \sin(\frac{3\theta}{7})$ (use the interval $0 \leq \theta \leq 14\pi$).

59. Without using technology, sketch the polar curve $\theta = \frac{2\pi}{3}$.

60. [T] Use a graphing utility to plot $r = \theta \sin \theta$ for $-\pi \leq \theta \leq \pi$.

61. [T] Use technology to plot $r = e^{-0.1\theta}$ for $-10 \leq \theta \leq 10$.

62. [T] There is a curve known as the “Black Hole.” Use technology to plot $r = e^{-0.01\theta}$ for $-100 \leq \theta \leq 100$.

63. [T] Use the results of the preceding two problems to explore the graphs of $r = e^{-0.001\theta}$ and $r = e^{-0.0001\theta}$ for $|\theta| > 100$.